VICTORY 2027

FORCE, WORK, POWER & ENERGY

LECTURE-1 PHYSICS

Shailendra sir



Topics To Be Covered

1. TYPES OF MOTION
2. MOMENT OF FORCE OR TORQUE
3. COUPLE
4. QUESTIONS


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Turns Into Topper
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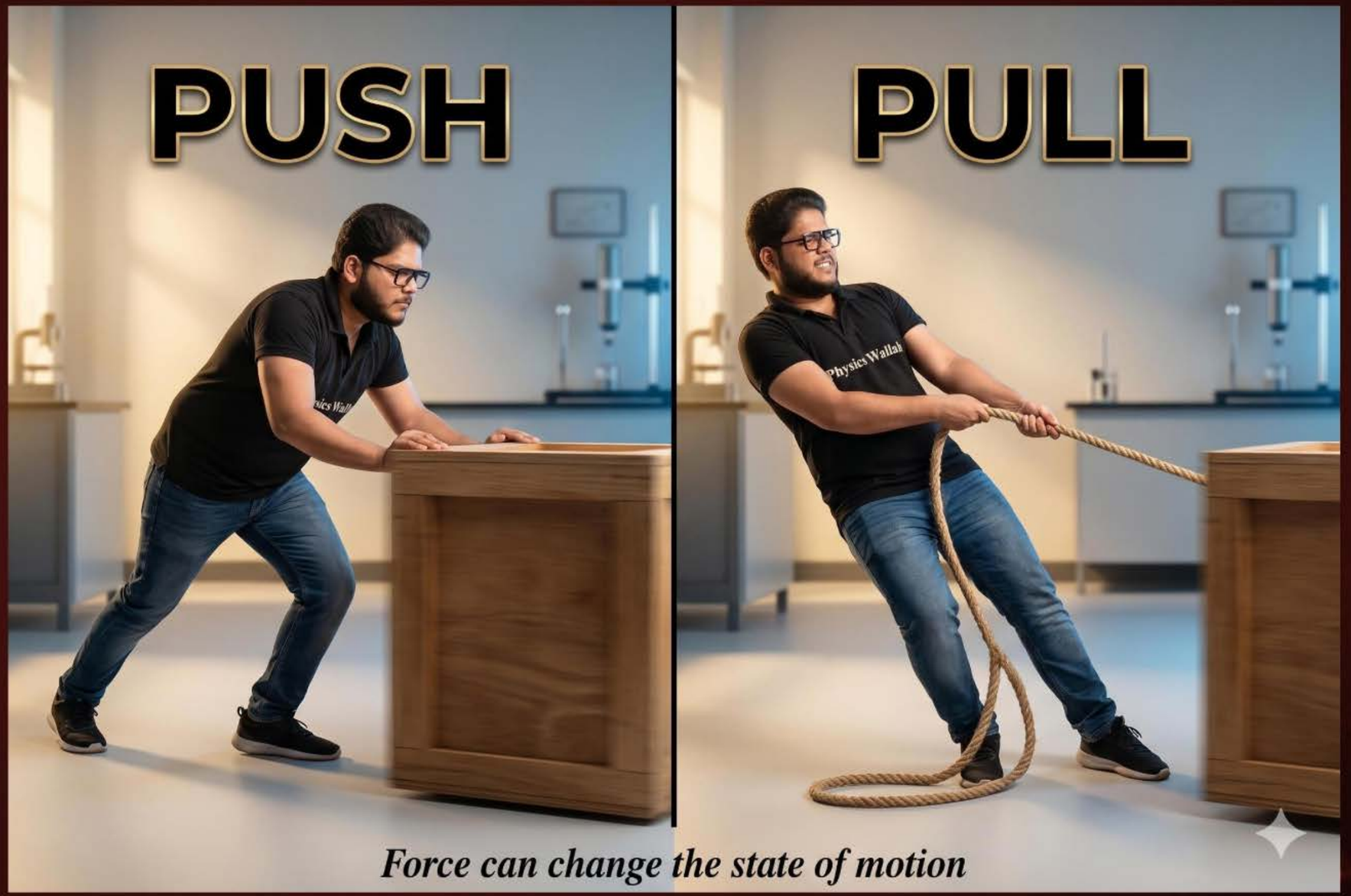
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FORCE

Selina

Push or pull applied for changing state of motion of any body is called force



EFFECTS OF FORCE

SSS OPI

A force applied on an object can produce four types of effects:

(1) Force can move a stationary object and can stop a moving object.

(2) Force can change speed of a moving object making it to move slower or faster.

(3) Force can change the direction of motion of an object.

(4) Force can change the shape of an object.

TYPES OF MOTION

A rigid body when acted upon by a force, can have two kinds of motion:

(1) linear or translational motion



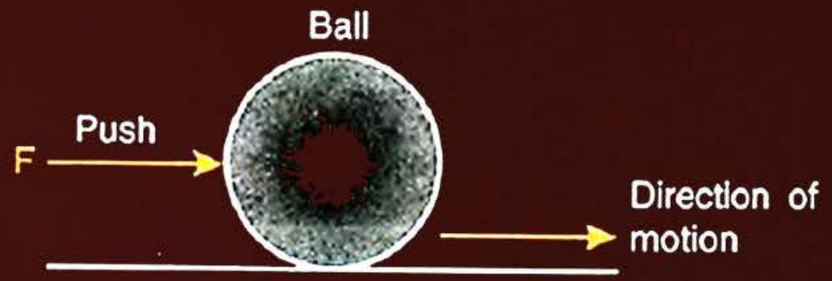
(2) rotational motion.



LINEAR OR TRANSLATIONAL MOTION

When a force acts on a stationary rigid body which is free to move, the body starts moving in a straight path in the direction of force. This is called the linear or translational motion.

For example, on pushing a ball lying on a floor, it begins to move.

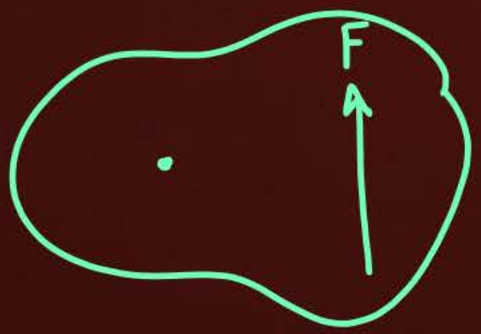


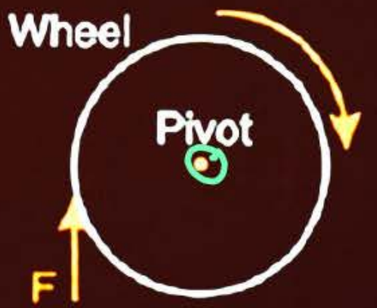
ROTATIONAL MOTION

If the body is pivoted at a point and the force is applied on the body at a suitable point, it rotates the body about the axis passing through the pivoted point. This is the turning effect of the force and the motion of body is called the rotational motion.

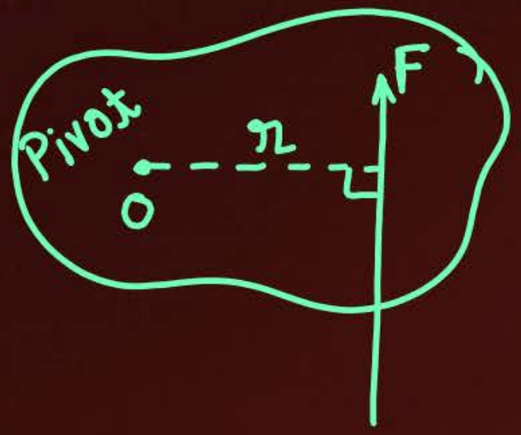
For example, if a wheel is pivoted at its centre and a force is applied tangentially on its rim, the wheel rotates about its centre.

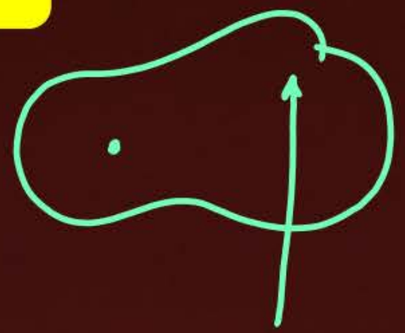
Similarly when a force is applied normally on the handle of a door, the door begins to rotate about an axis passing through the hinges on which the door rests.





MOMENT OF FORCE OR TORQUE

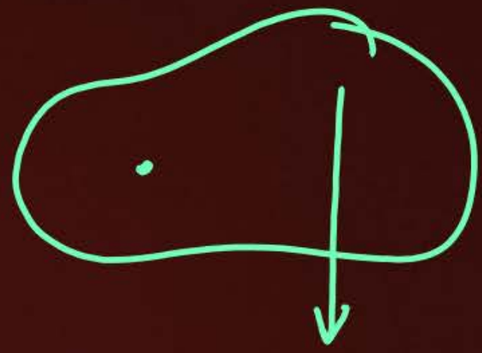




anticlockwise

$\tau \rightarrow$ outwards

(+ve)



Clockwise

$\tau \rightarrow$ Inwards

(−)

Moment of Force $= r \times F$

Vector Quantity

SI Unit : N m

CGS Unit : dyne cm

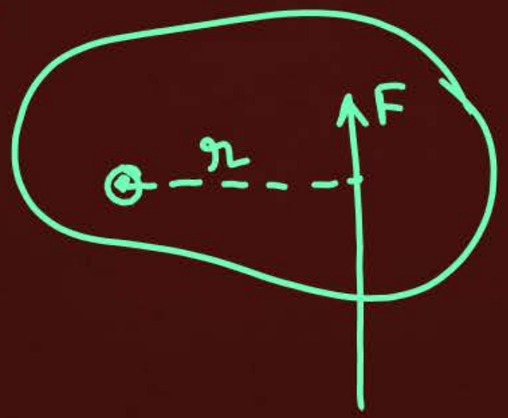
MOMENT OF FORCE OR TORQUE

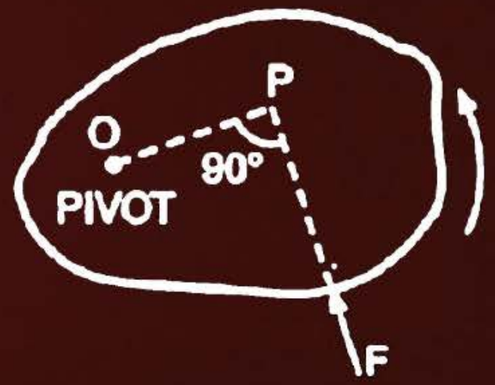
The turning effect on a body by a force depends on the following two factors:

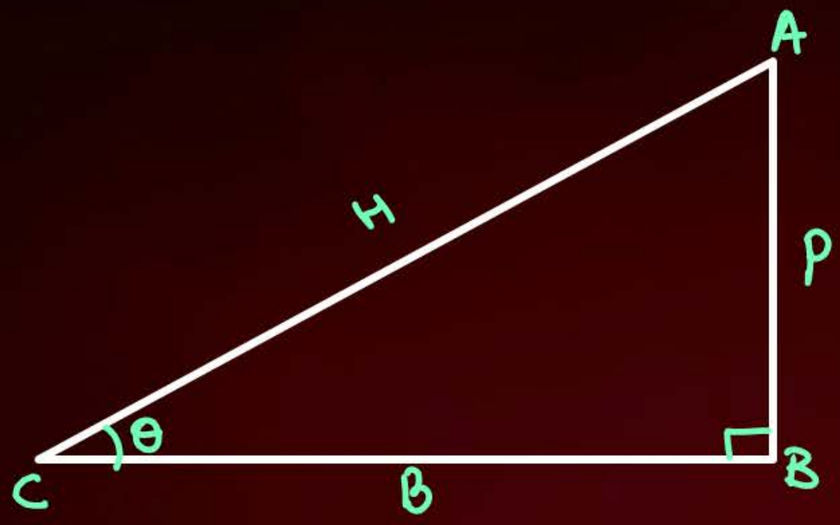
(1) the magnitude of the force applied,

(2) the distance of line of action of the force from the axis of rotation (or pivoted point).

$\tau = r \times F$



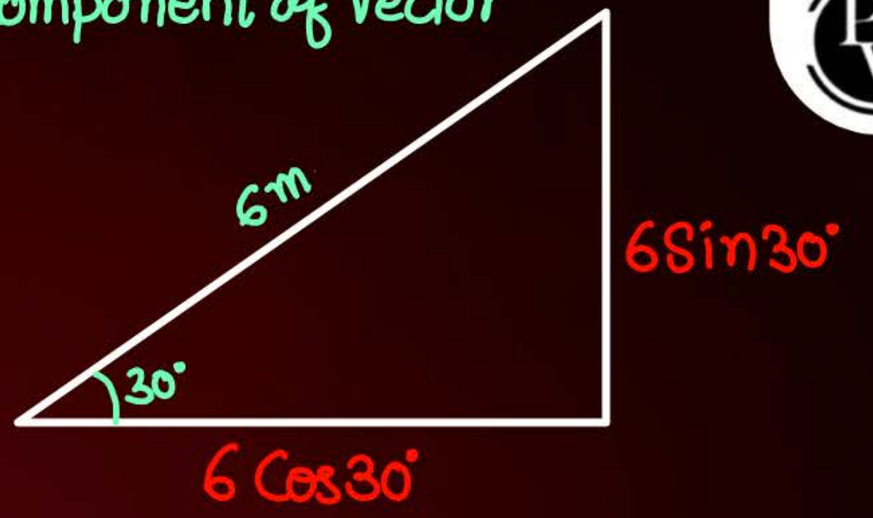


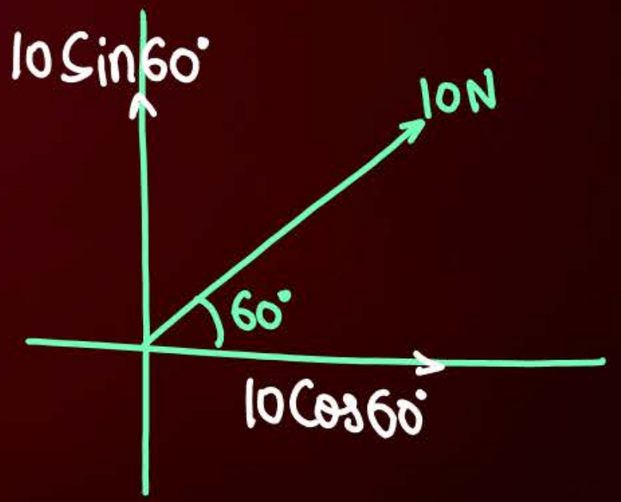


$$\sin\theta = \frac{AB}{AC} \Rightarrow \boxed{AB = AC\sin\theta}$$

$$\cos\theta = \frac{BC}{AC} \Rightarrow \boxed{BC = AC\cos\theta}$$

Component of Vector





MOMENT OF FORCE OR TORQUE

The turning effect on the body about an axis is due to the moment of force (or torque) applied on the body.

The moment of a force (or torque) is equal to the product of the magnitude of the force and the perpendicular distance of the line of action of force from the axis of rotation.

MOMENT OF FORCE OR TORQUE

Unit of moment of force = unit of force × unit of distance

The S.I. unit of force is newton and that of distance is metre, so the S.I. unit of moment of force is newton × metre. This is abbreviated as Nm.

The C.G.S. unit of moment of force is dyne × cm.

QUESTIONS

Moment of force is given by [SQP 2018]

1. force × parallel distance between axis of rotation.
2. force × perpendicular distance.
3. Torque × perpendicular distance.
4. None of these

QUESTIONS

A door lock is opened by turning the lever (handle) of length 0.2 m . If the moment of force produced is 1 Nm , then the minimum force required is **[MAIN 2024]**

$\tau = r \times F$

$1 = 0.2 \times F$

$F = \frac{1}{0.2} = 5\ N$

1. 5 N
2. 10 N
3. 20 N
4. 0.2 N

QUESTIONS

Calculate the force which will produce a moment of force of 1575 dyne-cm, when the perpendicular distance between point of application of force and turning point is 45 cm .

[SQP 2024]

1. 75 dyne
2. 45 dyne
3. 35 dyne
4. 25 dyne

$\tau = 1575$

$r = 45\ cm$

$F = ?$

$\tau = r \times F$

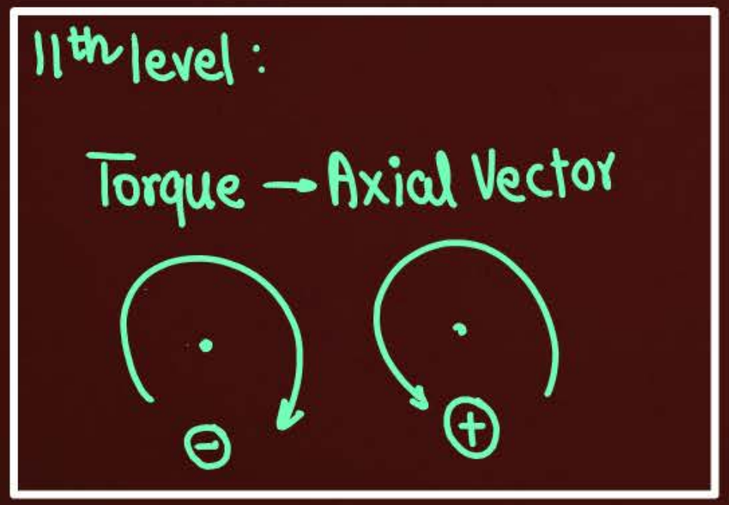
$1575 = 45 \times F$

$F = \frac{1575}{45} = 35\ dyne$

CLOCKWISE AND ANTICLOCKWISE MOMENTS

Conventionally, if the effect on the body is to turn it anticlockwise, moment of force is called the anticlockwise moment and it is taken positive, while if the effect on the body is to turn it clockwise, the moment of force is called the clockwise moment and it is taken negative.

The moment of force is a vector quantity. The direction of anticlockwise moment is along the axis of rotation outwards, while of clockwise moment is along the axis of rotation inwards.



QUESTIONS

A moment of couple has a tendency to rotate the body in an anticlockwise direction. Then the moment of couple is taken as:

1. positive
2. negative
3. maximum
4. zero

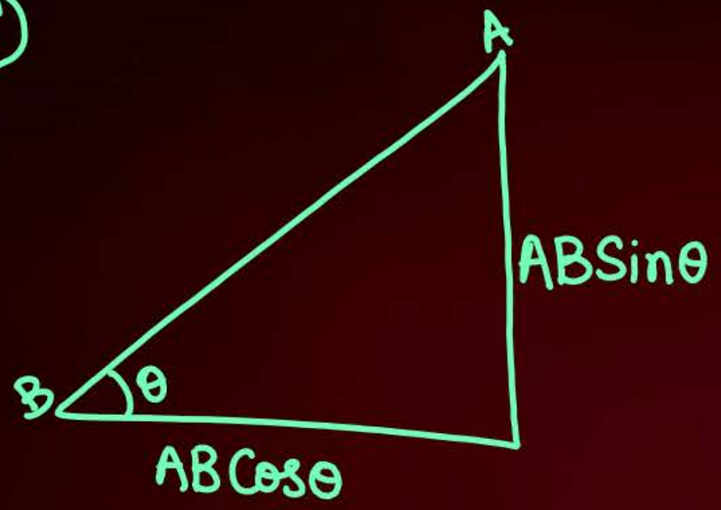
QUESTIONS

Clockwise moments are considered to be .....

[COMP 2012]

1. zero
2. positive
3. negative
4. all of these

SSS



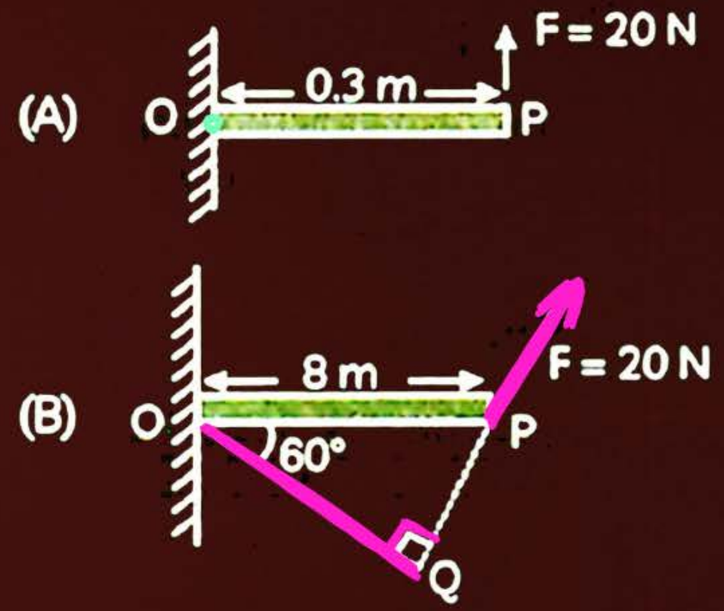
QUESTIONS

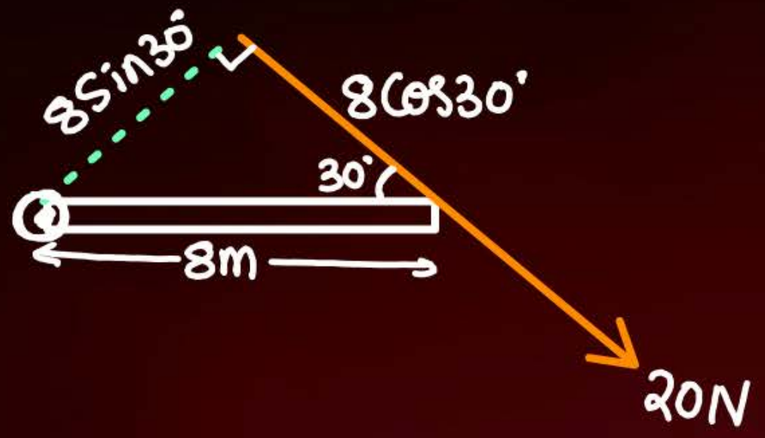
Find moment of force about O in the following.

(A) $\tau = r \times F = 0.3 \times 20 = 6\ Nm$

(B) $OQ = 8\cos 60^\circ = 8 \times \frac{1}{2} = 4\ m$

$\tau = r \times F = 4 \times 20 = 80\ Nm$





$$\tau = r \times F$$

$$= (8\sin 30^\circ) \times 20$$

$$= 8 \times \frac{1}{2} \times 20^{10}$$

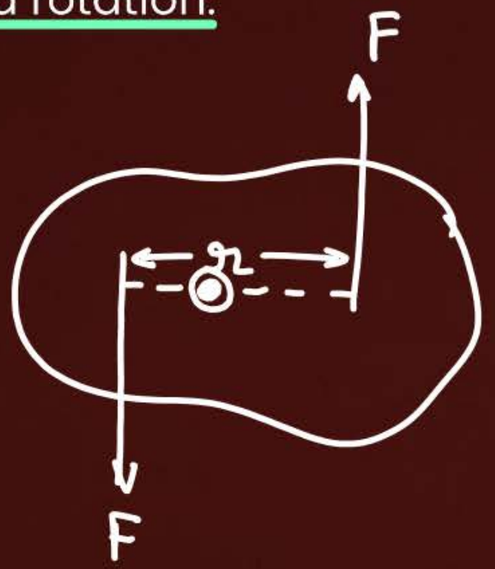
$$= 80 \text{ Nm (Clockwise)}$$

$$\tau = -80 \text{ Nm}$$

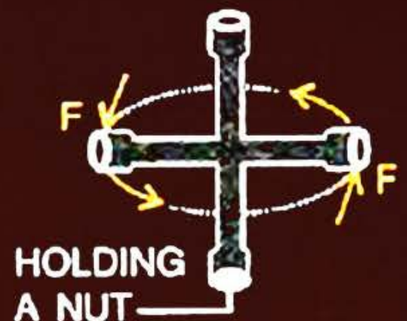
COUPLE

5→6:15 PM

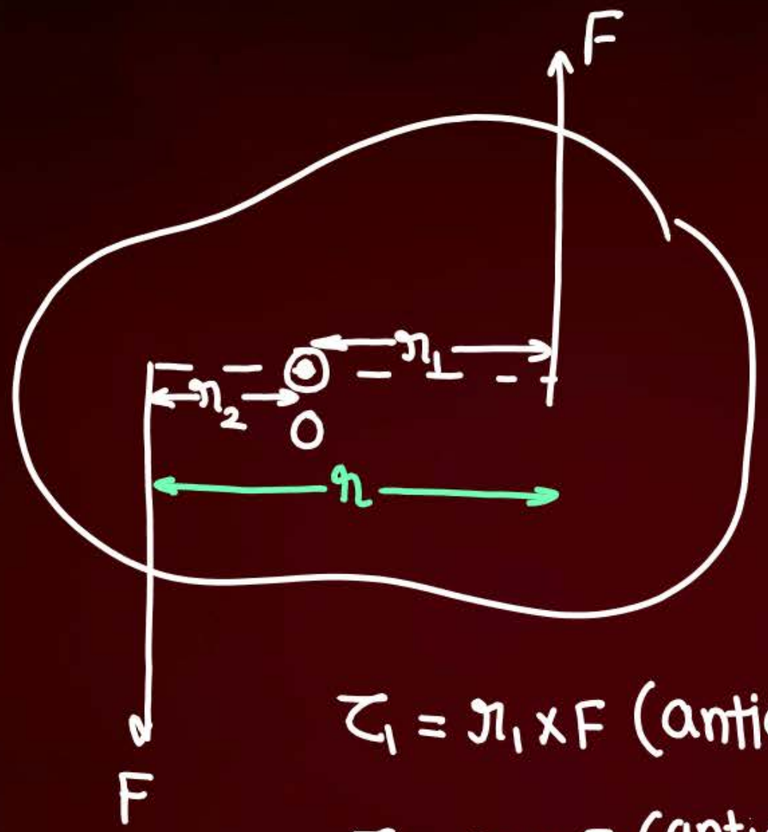
Two equal and opposite parallel forces, not acting along the same line, form a couple. A couple is always needed to produce a rotation.



$$\tau = r \times F$$



(a) Car wrench



$\tau_1 = r_1 \times F$ (anticlockwise)

$\tau_2 = r_2 \times F$ (anticlockwise)

$\tau_{total} = r_1 \times F + r_2 \times F$

$= (r_1 + r_2) F$

$\boxed{\tau = r \times F}$

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COUPLE

- Similarly turning a water tap (Fig. 1.10),
- tightening the cap of an inkpot (Fig. 1.11),
- turning the key in the hole of a lock (Fig. 1.12),
- turning the steering of a motor-car (Fig. 1.13), are the other examples where a pair of forces (couple) is applied for rotation.



Fig. 1.10 Turning a water tap



Fig. 1.11 Tightening the cap



Fig. 1.12 Turning a key in a lock



Fig. 1.13 Turning a steering wheel

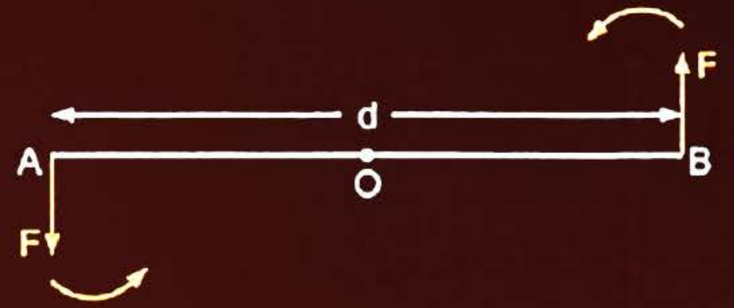
MOMENT OF COUPLE

Moment of force F at the end A

$$= F \times OA \text{ (anticlockwise)}$$

Moment of force F at the end B

$$= F \times OB \text{ (anticlockwise)}$$



Total moment of couple

$$= F \times OA + F \times OB$$
$$= F \times (OA + OB) = F \times AB$$
$$= F \times d \text{ (anticlockwise)}$$

In other words, the moment of couple is equal to the product of the either force and the perpendicular distance between the line of action of both the forces.

QUESTIONS

The moment of a force about a given axis depends:

$\vec{\tau} = \vec{r} \times \vec{F}$

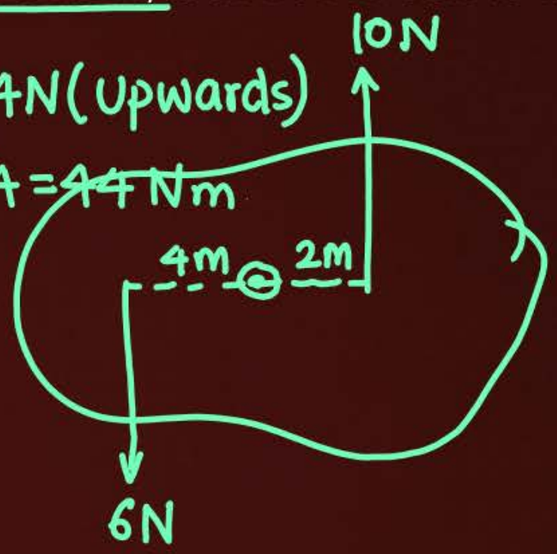
1. only on the magnitude of force
2. only on the perpendicular distance of force from the axis
3. neither on the force nor on the perpendicular distance of force from the axis
4. both, on the force and its perpendicular distance from the axis.

QUESTIONS

A body is acted upon by two unequal forces in opposite directions, but not in same line. The effect is that:

Linear / Translational ⇐ $F_{net} = 10 - 6 = 4N$ (Upwards)

Rotational ⇐ $\tau_{net} = 10 \times 2 + 6 \times 4 = 44\ Nm$



1. the body will have only the rotational motion
2. the body will have only the translational motion
3. the body will have neither the rotational motion nor the translational motion
4. the body will have rotational as well as translational motion.

QUESTIONS

HW

A body is pivoted at a point. A force of $10\,\mathrm{N}$ is applied at a distance of $30\,\mathrm{cm}$ from the pivot. Calculate the moment of force about the pivot.

1. 1 Nm
2. 2 Nm
3. 3 Nm
4. 4 Nm

QUESTIONS

The moment of a force of 5 N about a point P is 2 Nm. Calculate the distance of point of application of the force from the point P.

$F = 5\,N$

$\tau = 2\,Nm$

$r \times F = 2$

$r \times 5 = 2$

$r = \frac{2}{5} = 0.4\,m$

1. 0.1 m
2. 0.2 m
3. 0.3 m
4. 0.4 m

QUESTIONS

Imp A mechanic can open a nut by applying a force of $150\ \text{N}$ while using a lever handle of length $40\ \text{cm}$. How long handle is required if he wants to open it by applying a force of only $50\ \text{N}$?

1. 1.1 m
2. 1.2 m
3. 1.3 m
4. 1.4 m

$F_1 = 150\text{N}$ $\quad F_2 = 50\text{N}$

$r_1 = 40\text{cm}$ $\quad r_2 = ?$

Required Torque

$= r_1 \times F_1$

$= \frac{4\phi}{100} \times 15\phi$

$= 60\ \text{Nm}$

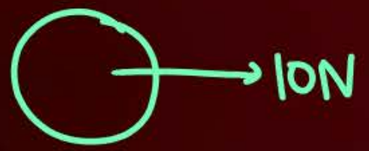
$\tau = r_2 \times F_2$

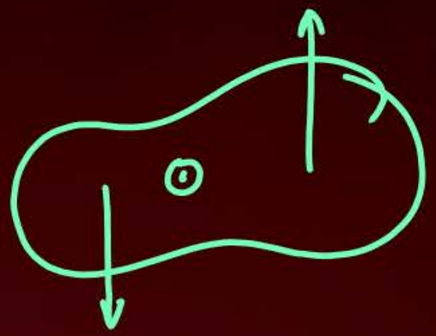
$60 = r_2 \times 50$

$r_2 = \frac{6\phi}{5\phi} = 1.2\ \text{m}$

Linear Motion → F







Rotational Motion → τ

QUESTIONS

A mechanic can open a nut by applying a force of $150\ \mathrm{N}$ while using a lever handle of length $40\ \mathrm{cm}$. How long handle is required if he wants to open it by applying a force of only $50\ \mathrm{N}$?

Solution:

In the first case, $\mathrm{F} = 150\mathrm{N}, r = 40\mathrm{cm} = 0.4\ \mathrm{m}$ The moment of force needed to open the nut

$$= 150\ \mathrm{N} \times 0.4\ \mathrm{m} = 60\ \mathrm{Nm}$$

In the second case, $\mathrm{F} = 50\mathrm{N}$,

If he uses the handle of length Lm, then

Moment of force $= 50N \times L\ m = \boldsymbol{50L\ Nm}$

From eqns. (i) and (ii),

$$50\mathrm{L} = 60$$

$$\mathrm{L} = \frac{60}{50} = 1.2\mathrm{m}$$

QUESTIONS

HW

The iron door of a building is 3 m broad. It can be opened by applying a force of 100 N normally at the middle of the door. Calculate:

(a) the torque needed to open the door,

(b) the least force and its point of application to open the door.

1. 100 Nm & 50 N
2. 150 Nm & 50 N
3. 150 Nm & 25 N
4. 100 Nm & 25 N

QUESTIONS

The iron door of a building is 3 m broad. It can be opened by applying a force of 100 N normally at the middle of the door. Calculate:

(a) the torque needed to open the door,

(b) the least force and its point of application to open the door.

Solution:

(a) $\mathrm{F} = 100\mathrm{N}$, distance of point of application of force, $r = \frac{1}{2} \times$ breadth of door $= \frac{1}{2} \times 3\mathrm{m} = 1.5\mathrm{m}$

Moment of the force needed to open the door

$$= \mathrm{F} \times r = 100\mathrm{N} \times 1.5\mathrm{m} = 150\mathrm{Nm}$$

(b) The force required will be least if it is applied at the farthest point from the hinges. Therefore the force should be applied at the free end of the door. i.e., at distance of $3\mathrm{m}$ from the hinges. Let the required force be F' newton, then

$$\text{Moment of force } = \mathrm{F}' \times 3\mathrm{Nm}$$

$$\mathrm{F}' \times 3 = 150$$

$$\mathrm{F}' = \frac{150}{3} = 50\mathrm{N}$$

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

THANK YOU SSS ARMY

FOR YOUR UNWAVERING SUPPORT & TRUST



our physics journey has JUST BEGUN.

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Thank You